

Project ID:

R26-SE-002

1. Topic (12 words max)

AI Ecosystem for Fake Job Scam Detection and Fraud-Aware Job Recommendation

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

- Online recruitment platforms have become a primary means of finding jobs due to their accessibility and reach among students, graduates, and unemployed individuals. However, this transformation has intensified online recruitment fraud, where fake postings exploit applicants for financial gain and identity theft.
- Studies report a significant increase in fraudulent advertisements, highlighting the severity of this problem [1], [2]. In developing countries like Sri Lanka, high demand for foreign and private-sector jobs makes the impact more critical due to limited verification mechanisms and digital literacy gaps.
- Existing research focuses on detecting fake postings using machine learning and NLP (Natural Language Processing) on job descriptions and structured metadata.
- Approaches using TF-IDF, ensemble classifiers (Random Forest, boosting), and balancing techniques (SMOTE, ADASYN) show high accuracy [1], [3], [4]. Public datasets like EMSCAD are widely used to improve detection [3], [5]. However, these solutions treat fraud as a single-stage classification problem, relying mainly on textual patterns within advertisements.
- In reality, recruitment fraud progresses beyond the posting stage, involving fake employers, impersonated identities, and deceptive communications post-application.
- This research addresses this gap by modeling fraud as a multi-stage process via an AI ecosystem integrating job detection, employer legitimacy verification, communication analysis, and user-centric intelligent job recommendation and ranking system with fraud-aware personalization.

- By verifying employer identity through domain consistency, analyzing communications for fraud patterns, and recommending with ranking the best job option through real ones the system provides more comprehensive protection than single-model approaches.
- This research aligns with the **Artificial Intelligence Machines and Systems (AIMS) cluster**, targeting **HR technology**.
- Socially and economically, it aims to reduce financial losses, protect data, and enhance trust. Furthermore, the project contributes to **UN SDG 8 (Decent Work and Economic Growth)** and **SDG 16 (Peace, Justice, and Strong Institutions)** by combating digital fraud and exploitation in recruitment systems. Specifically, it empowers users to verify recruiters before sharing sensitive information, closing the security gap in modern labor markets.

References:

- [1] M. Allam *et al.*, “Comprehensive Detection of Fake Job Postings: Applying Advanced Data Balancing and Machine Learning Techniques to Combat Online Frauds,” *Proc. 6th Int. Conf. Recent Advances in Information Technology (RAIT)*, IEEE, 2025.
<https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=11089425>
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<https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10837920>
- [3] J. Medapati *et al.*, “Detection of Fake Online Recruitment using Machine Learning Approach,” *Proc. 3rd Int. Conf. Augmented Intelligence and Sustainable Systems (ICAISS)*, IEEE, 2025. <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=11041922>
- [4] B. Alghamdi and F. Alharby, “An Intelligent Model for Online Recruitment Fraud Detection,” *Journal of Information Security*, vol. 10, no. 3, pp. 155–176, 2019.
https://www.researchgate.net/publication/334365773_An_Intelligent_Model_for_Online_Recruitment_Fraud_Detection#fullTextFileContent
- [5] N. S. Kaggle Notebook, “CV Project v0.1 – Fake Job Postings,” Kaggle, 2023. [Online]. Available: <https://www.kaggle.com/code/nithin1729s/cv-project-v01>
2025. <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=11041922>

6. Brief description of Existing Research and Systems (300 words max)

- Existing solutions and research related to fake job scam detection primarily focus on identifying fraudulent job advertisements using machine learning and natural language processing techniques.
- Several online recruitment platforms apply basic moderation mechanisms such as manual review, keyword filtering, and user reporting to reduce scam job postings. However, these systems are often reactive, rely heavily on human intervention, and struggle to scale with the rapidly increasing volume of online job advertisements.
- Recent academic research has explored automated fake job post detection using supervised machine learning models. Studies such as Medapati et al. (2025) demonstrate the effectiveness of NLP-based classifiers by analyzing job descriptions, salary information, company profiles, and metadata to distinguish between real and fake job postings. Similarly, Vijayalakshmi et al. (2024) propose Python-based analytical frameworks that leverage structured and unstructured job data to classify fraudulent recruitment posts. Public datasets, such as the Kaggle Employment Scam Aegean dataset, have further supported research by providing labeled job advertisement data for experimentation and benchmarking.
- Despite these advancements, existing research exhibits several **critical limitations**. Most studies restrict their analysis to the **pre-application stage**, focusing only on job advertisement content. They do not verify the legitimacy of the employer behind the job post, allowing scammers to impersonate real companies using realistic descriptions.
- Furthermore, existing systems fail to analyze **recruiter–candidate communication**, where scams frequently occur through requests for money, personal documents, or urgency-based manipulation via emails and messaging platforms.
- Another major limitation is the lack of **explainability**, as many systems provide only binary outputs without clear reasoning, reducing user trust.
- These gaps highlight the need for a comprehensive, multi-stage solution that goes beyond job post classification. Addressing employer legitimacy, post-application communication scams, and recommend and rank the best job option remains largely unexplored in current research, motivating the proposed multi-model AI-based system.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

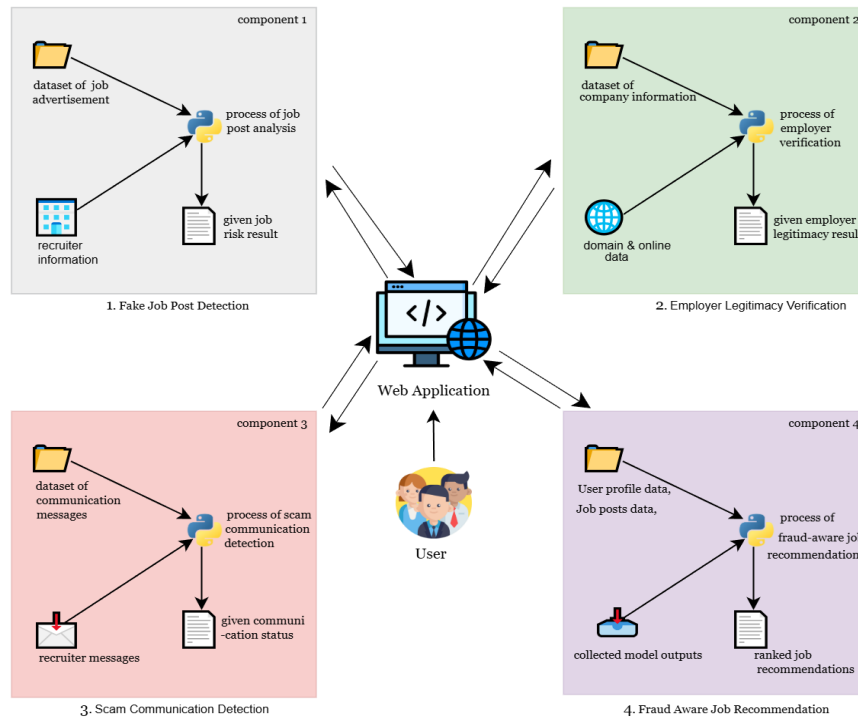


Diagram URL:

<https://drive.google.com/file/d/1jYdrFSrQpjU7yH8BgtHCn1JvfK5Zzg1A/view?usp=sharing>

The proposed research introduces an **AI-driven, end-to-end fraud-aware job intelligence system** designed to protect job seekers from fake job scams while simultaneously improving the quality and safety of job recommendations. This solution directly addresses the critical gaps identified in existing systems, which largely focus on job–skill matching and text-based fraud detection in isolation, without validating employer legitimacy, analyzing scam communications, or integrating risk awareness into job recommendations.

Existing job platforms and research systems typically operate in disconnected way—detecting fake job posts using NLP techniques or recommending jobs based solely on user preferences. However, they fail to; verify the authenticity of employers, detect scam behavior during recruiter–candidate communication, and incorporate fraud risk into personalized job recommendations. The proposed system bridges these gaps by integrating **four interconnected AI components** within a unified web-based platform, enabling holistic fraud detection and safer decision-making for job seekers.

Inputs

The system accepts multiple heterogeneous inputs through a centralized web application:

- **Job Advertisement Data** (job title, description, salary, location, recruiter email)
- **Company and Employer Data** (company name, domain information, online presence)
- **Recruiter Communication Data** (emails, chat messages)
- **User Profile Data** (skills, education, experience, job preferences)

Process

The solution operates through four coordinated AI-driven components:

1. **Fake Job Post Detection (Component 1)**

This module analyzes job advertisements using NLP and metadata-based techniques to identify unrealistic salaries, spam phrases, and suspicious recruiter details. It produces a **job risk score** indicating whether a job post is fake, suspicious, or legitimate.

2. **Employer Legitimacy Verification (Component 2)**

This component evaluates employer credibility by analyzing company websites, domain age, LinkedIn presence, and consistency of digital footprints. A novel **Employer Completeness Score** is generated to classify employers as legitimate, suspicious, or fake with clear specific 'flags' to explain exactly why an employer was flagged.

3. **Scam Communication Detection (Component 3)**

Recruiter messages are analyzed to detect scam patterns such as urgency pressure, payment requests, or credential harvesting. The model outputs a **communication risk status**, indicating the likelihood of scam behavior.

4. **Fraud-Aware Job Recommendation (Component 4)**

This final component aggregates outputs from the first three models along with user profile data. It applies a fraud-aware ranking strategy that balances **skill match relevance** with **fraud risk penalties**, generating **ranked job recommendations** (Best, Second Best, Third Best) along with explainable justifications.

Outputs

- Job post fraud risk classification
- Employer legitimacy score
- Communication scam status
- **Ranked, fraud-aware job recommendations** with explanation
- Overall recommendation score (0–100)

From a system perspective, the proposed solution is **modular, explainable, and user-centric**, enabling independent model development while supporting seamless integration. By embedding fraud awareness into every stage of the job-seeking lifecycle, the system moves beyond detection toward **proactive prevention and intelligent guidance**, offering a practical and scalable solution to modern job scam challenges.

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

- This research requires interdisciplinary expertise across Artificial Intelligence, Natural Language Processing (NLP), Cybersecurity, and Online Fraud Detection domains.
- Strong knowledge of machine learning classification techniques, including TF-IDF, n-gram analysis, logistic regression, decision trees, and clustering algorithms (e.g., K-Means), is essential for building and evaluating the proposed multi-model ecosystem. Additionally, expertise in explainable AI (XAI) is required to generate transparent, user-friendly risk explanations and actionable warnings.
- Domain knowledge in online recruitment processes, employment fraud patterns, and digital scam behaviors is critical.
- This includes understanding common fake job indicators such as unrealistic salaries, informal communication styles, free email domains, identity theft attempts, advance-fee scams, and social engineering tactics used in scam communications. Knowledge of web and domain legitimacy heuristics, such as domain age, website consistency, and email-domain matching, is also required for employer verification.
- The system relies heavily on text-based data, requiring proficiency in data preprocessing, feature engineering, and anomaly detection for unstructured textual content. Familiarity with ethical AI practices, data privacy, and responsible handling of sensitive information (e.g., personal details in scam communications) is necessary.
- The data requirements include job advertisements (real and fake), employer metadata (company names, websites, email domains), and scam communication datasets (emails, chat messages).
- Data may be sourced from public job portals, cybersecurity forums, scam-report databases, and synthetically generated samples to address class imbalance. Labeled datasets are required for supervised learning, while unlabeled data supports clustering and scam pattern discovery.
- This combined expertise enables accurate detection, verification, explanation, and prevention of job-related scams within a unified AI ecosystem.

9. Objectives and Novelty

Main Objective The aim of this research is to design and develop an end-to-end, multi-model AI-based system that detects fake job scams across the entire recruitment lifecycle by analyzing job advertisements, verifying employer legitimacy, monitoring recruiter–candidate communication, and providing explainable risk assessments to protect job seekers from online employment fraud.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Kajanthan U IT22224002	Scam Communication Detection Model	Real World Problem: After a job seeker applies for a job, communication usually continues through email, WhatsApp, or chat platforms . While many existing systems detect fake job advertisements , scammers often succeed after this stage by sending deceptive messages to job seekers. This function focuses on detecting scam behavior during recruiter–candidate communication. Inputs: - Email messages - WhatsApp messages - Chat conversations related to recruitment	The Scam Communication Detection model introduces a quantitative, behavior-based scoring system for recruiter–candidate messages, rather than a simple yes/no classification. It evaluates multiple scam indicators and assigns risk levels to reflect the degree of fraudulent intent . Key Features / Why Novel: Post-Application Communication Focus: Detects scams in email, WhatsApp, and chat messages after job application, a stage overlooked by existing research. Behavior-Based Scam Detection: Identifies actions like

		Model Outputs: - Scam / Legitimate - Risk level (Low / Medium / High) - Explanation of detected scam indicators Task 1: Requirement Analysis & Research Gap Identification - Review existing fake job detection literature (e.g., Medapati et al., 2025) - Identify limitations related to post-application scam detection - Analyze real-world scam communication patterns via email and messaging platforms - Define the scope of communication channels (email, WhatsApp, chat) Outcome: Clearly defined research gap and functional requirements for communication-level scam detection Task 2: Data Collection & Dataset Construction - Collect sample recruitment-related messages (legitimate and scam)	money requests, personal document requests, and urgency/pressure tactics. Multi-Indicator & Combined Pattern Detection: Combines multiple scam signals; higher risk is assigned when multiple indicators appear together (e.g., money + urgency). Hybrid Rule-Based + Machine Learning Approach: Integrates pattern/rule detection with ML classification for both accuracy and explainability. Output: - Communication Classification: Scam / Legitimate - Risk Level: Low / Medium / High - Explainable indicators: Money request, Document request, Urgency/Pressure detected
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		• Create privacy-safe datasets by anonymizing sensitive information • Generate synthetic scam messages to simulate diverse scam scenarios: ○ Money requests ○ Document requests ○ Urgency-based manipulation • Balance scam and legitimate message samples Outcome: Hybrid dataset containing real and synthetic recruiter–candidate messages Task 3: Text Preprocessing & Cleaning • Normalize message text (lowercasing, punctuation removal) • Remove stop words and irrelevant symbols • Handle emojis, abbreviations, and informal chat language • Tokenize text for feature extraction Outcome: Cleaned and standardized communication dataset	
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		Task 4: Feature Engineering & Scam Indicator Extraction • Apply TF-IDF vectorization to represent message text numerically • Design keyword-based binary indicators: ○ money_flag ○ doc_flag ○ urgency_flag • Implement regex-based pattern detection: ○ Bank account numbers ○ OTP formats ○ Payment references • Detect mixed scam indicators (e.g., money + urgency) Outcome: Rich feature set combining textual, rule-based, and pattern-based indicators Task 5: Model Development & Training • Train binary classification models: ○ Logistic Regression	
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		○ Random Forest ○ Support Vector Machine (SVM) • Combine TF-IDF vectors with indicator flags as model inputs • Tune hyperparameters for optimal performance Outcome: Trained Scam Communication Classification Model Task 6: Rule + Machine Learning Hybrid Decision Logic • Implement rule-based checks for high-risk patterns • Fuse rule-based outputs with ML prediction scores • Design a Risk Score (0–100) calculation strategy • Map scores to risk levels: Low / Medium / High Outcome: Hybrid detection logic with explainable risk scoring Task 7: Model Evaluation & Performance Validation • Evaluate model using:	
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		○ Accuracy ○ Precision ○ Recall ○ F1-score • Analyze false positives and false negatives • Validate robustness against unseen scam message patterns Outcome: Verified and reliable scam detection performance metrics Task 8: Explainability & Output Generation • Generate user-friendly explanations for detections • Identify detected scam indicators (money, documents, urgency) • Structure output in JSON format for easy integration Outcome: Transparent and interpretable detection results Task 9: Real-Time Integration & System Deployment • Integrate model with backend service	
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		• Enable real-time message analysis and alert generation • Support multi-channel inputs (email, WhatsApp, chat) Outcome: Real-time scam detection capability within the system ecosystem	
Santhosh S IT22113740	Fake Job Post Detection Model (ML + NLP)	Real World Problem: Online job portals and social media platforms are increasingly used to advertise job vacancies. However, scammers frequently publish fake job advertisements that promise high salaries, easy work, or urgent hiring to attract job seekers. These fake posts are often the first entry point of recruitment scams and are responsible for misleading users before any employer verification or communication takes place. Job seekers—especially students, fresh graduates, and unemployed individuals—find it difficult to distinguish between genuine and fraudulent job advertisements due to their realistic appearance. Inputs: • Job title • Job description text • Salary information	Fraud Risk Scoring System • Convert model probability Risk Score (0–100) • Define risk levels: • 0–30 → Low • 31–70 → Medium • 71–100 → High • Output passed to Model 4

		• Job location • Employment type • Recruiter email address • Company name (optional metadata) Model Outputs: • Job Classification ○ Fake ○ Suspicious ○ Legitimate • Fraud Risk Level ○ Low Risk ○ Medium Risk ○ High Risk • Optional Indicators (for explanation) ○ Unrealistic salary detected ○ Free email domain used ○ Spam keywords present ○ Incomplete job description Task 1: Dataset Selection & Preparation • Select benchmark fake job posting datasets	
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		- Identify text and metadata fields - Clean and preprocess job advertisement data Outcome: Structured dataset for training Task 2: Text Preprocessing & Feature Extraction - Apply tokenization, stop-word removal, and normalization - Extract TF-IDF and n-gram features from job title and description - Detect spam-related keywords and phrases Outcome: Numerical representation of job content Task 3: Metadata Anomaly Detection - Analyze salary–role mismatches - Identify unrealistic salary ranges - Detect free vs corporate email domains - Check missing or incomplete job information Outcome: Metadata-based fraud indicators Task 4: Model Training & Classification	
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		• Train ML classifiers (Logistic Regression, Random Forest, SVM) • Combine text features and metadata features • Optimize model performance Outcome: Trained fake job detection model Task 5: Risk Scoring & Label Assignment • Convert model outputs into risk scores • Define thresholds for Fake / Suspicious / Legitimate • Assign fraud risk levels (Low / Medium / High) Outcome: Interpretable classification results Task 6: Model Evaluation & Validation • Evaluate using accuracy, precision, recall, and F1-score • Perform error analysis on misclassified posts • Validate robustness against unseen job advertisements Outcome: Verified and reliable model performance	
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		Task 7: Integration & Output Generation - Integrate Model 1 into the multi-model ecosystem - Provide outputs to: - Model 2 (Employer Verification) - Model 4 (Risk Explanation) Outcome: Seamless pipeline integration	
Bandara H M N T IT22251664	Employer (Company) Legitimacy Verification Model	Real World Problem: Many scams happen when job post looks legitimate, but: The company is fake, email domain is impersonated, website is newly created or copied. Inputs: - Company Name: Used to verify: - Does this name exist in business directories? - Does this name match the website? - Website URL: Check: - Is this a valid corporate site?	- The core novelty of this model is the Employer Completeness Score. Instead of a simple 'yes or no' binary decision, we use a quantitative measure from 0 to 100. - This score represents how professional and consistent an employer's digital footprint is across the web. - We classify the results into three tiers: 71-100 (Legitimate): A strong, consistent online presence. 41-70 (Suspicious): Partial or inconsistent

		- Is it a suspicious site made with free hosting? 3. Email Domain (from job post): Examples: - hr@dialog.lk -> Legit - dialoglkjobs@gmail.com -> Suspicious - infosys-support@india-careers.info -> High risk 4. Domain Age (WHOIS data): Scammers usually buy new domains. - Domain age < 6 months -> Suspicious - Domain age < 1 month -> High risk 5. Logo & Company Description Consistency: Fake sites reuse stock logos or mismatched logos. 6. Location + Address Verification: Check mismatch: - Job says “Colombo office” - Website address: “Nigeria / UK” 7. Social Media Existence: Real companies have: - LinkedIn page	information that requires caution. 0-40 (Fake): A fragmented or missing digital footprint. - The final output doesn't just give a score; it provides specific 'flags'—like 'Email domain does not match' or 'No LinkedIn presence'—to explain exactly why an employer was flagged." - And the output is passed to Model 4
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		• Facebook page • Google reviews Model Outputs: 1. Employer Risk Score (0–100) • 0–40 → Fake • 41–70 → Suspicious • 71–100 → Legit 2. Flags + Explanations • “Email domain does not match official website.” • “Domain registered only 23 days ago.” • “No company presence on LinkedIn.” • “Website logo does not match company name.” • “Website hosted on free hosting platform.” 3. Final Classification • Legitimate Company • Suspicious Employer • Fake Employer Task 1: Requirement Analysis & Research Review	
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		• Study existing fake job detection approaches and identify limitations related to employer verification • Analyze how employers are impersonated in real-world job scams • Identify employer-related attributes needed for legitimacy verification Outcome: Clear understanding of research gap and feature requirements Task 2: Dataset Selection & Employer Feature Identification • Select suitable benchmark datasets (e.g., Kaggle fake job postings dataset) • Identify employer-specific fields such as company name, profile, logo, and contact information • Define features required to compute employer completeness Outcome: Structured employer feature list for dataset creation Task 3: Employer Feature Engineering	
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		• Extract and preprocess employer-related attributes • Engineer new features such as: ○ Website availability indicator ○ Email domain type (corporate vs free) ○ Company profile completeness score ○ Logo presence indicator ○ Contact detail consistency Outcome: Employer-centric feature dataset Task 4: Employer Completeness Score Design (Core Novel Task) • Design a weighted scoring mechanism to compute Employer Completeness Score (0–100) • Normalize feature values for fair scoring • Define risk thresholds (Fake / Suspicious / Legit) Outcome: Explainable and interpretable trust scoring framework	
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		Task 5: Model Development & Classification • Use machine learning or rule-based logic to classify employers based on completeness score • Implement interpretable models (Decision Tree / Logistic Regression) • Ensure explainability of decisions Outcome: Functional Employer Legitimacy Verification Model Task 6: Model Evaluation & Validation • Evaluate model using accuracy, precision, recall, and confusion matrix • Validate score behavior using real and scam employer samples • Perform error analysis on misclassified cases Outcome: Performance metrics and validated scoring logic Task 7: Integration with System Architecture • Integrate Model 2 outputs with:	
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		○ Model 1 (Job Post Detection) ○ Model 4 (Risk Explanation) • Provide employer trust score and risk label as output Outcome: Seamless integration within multi-model ecosystem	
Methmi N P C S IT22188618	Fraud-Aware Job Recommendation Model	Real World Problem: Existing job recommendation systems focus mainly on skill matching and user preferences while ignoring fraud risks, which can lead job seekers to be recommended suitable but unsafe jobs that result in scams and financial loss. Inputs: User Profile Inputs • User skills • Educational background • Work experience level • Preferred job type (full-time / part-time / internship) • Preferred job location Job Posting Inputs	• This system introduces a ranked job recommendation mechanism that presents job opportunities in a prioritized order rather than as an unstructured list. • Instead of simply labeling jobs as safe or unsafe, the model intelligently evaluates multiple factors—such as job relevance to the user profile, employer legitimacy score, job post risk level, and communication safety indicators—and combines them to generate a fraud-aware ranking score. • Based on this score, job opportunities are categorized and displayed as:

		• Job title • Job description • Salary information • Job location • Company name • Recruiter email domain • Job posting age 3. Risk Indicators (synthetic or simulated) • Job content risk score • Employer risk score • Communication risk score (synthetic or simulated) Model Outputs: 1. Ranked Job Recommendations • Best Option • Second Best Option • Third Best Option 2. Overall Recommendation Score (0–100) • Skill Match: 85 • Fraud Risk Penalty: 25 • Overall Recommendation Score: 60 3. Explainable Recommendation Justification	○ Best Option – Highly relevant to the user, verified employer, and lowest fraud risk ○ Second Best Option – Moderately relevant with minor risk indicators ○ Third Best Option – Acceptable relevance but higher caution level • This ranking approach is novel because most existing recruitment platforms either recommend jobs purely based on user preferences or simply block suspicious jobs. By integrating fraud risk signals into the recommendation process, the system not only suggests suitable jobs but also guides users toward safer choices, enabling informed decision-making.
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		• “Score reduced due to free email domain usage and unrealistic salary range.” • “Score reduced due to missing company website and newly created recruiter email domain.” Task 1: Requirement Analysis & Research Gap Identification • Study existing job recommendation systems and fake job detection literature • Identify limitations of traditional recommender systems that ignore fraud risk • Analyze real-world job scam scenarios where suitable jobs are unsafe • Define functional and non-functional requirements for fraud-aware recommendations Outcome: Clearly defined research gap and system requirements for fraud-aware job recommendation Task 2: Dataset Selection & Preparation	
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		• Select publicly available job posting datasets (e.g., Kaggle job datasets) • Define user profile schema for recommendation • Clean and structure job and user data • Create synthetic user profiles for experimentation Outcome: Structured datasets for job postings and user profiles Task 3: Skill Matching & User Profiling • Extract skill keywords from job descriptions • Match user skills with job requirements • Compute Skill Match Score (0–100) • Normalize user preference weights (location, job type) Outcome: Quantified skill match and user suitability scores Task 4: Fraud Risk Simulation & Feature Engineering • Design proxy fraud indicators (Spam keyword density, Unrealistic	
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		salary detection, Free email domain usage, Incomplete job information) • Simulate communication-related risk patterns • Normalize individual risk indicators to a common scale Outcome: Simulated fraud risk feature set independent of other models Task 5: Fraud-Aware Recommendation Logic Design • Design weighted fusion strategy for multiple risk indicators • Develop final recommendation scoring formula (Skill match reward, Fraud risk penalty) • Define thresholds for Low / Medium / High risk jobs Outcome: Mathematically defined fraud-aware ranking framework Task 6: Model Development & Ranking Implementation • Implement recommendation logic using rule-based or ML-based ranking	
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		- Rank jobs for each user profile - Identify top recommendations and unsafe jobs Outcome: Functional fraud-aware job recommendation engine Task 7: Evaluation & Performance Analysis - Compare fraud-aware recommendations with traditional recommendations - Measure reduction of high-risk jobs in top recommendations - Conduct case-based analysis using simulated scam scenarios Outcome: Validated improvement in recommendation safety and reliability Task 8: Explainability & Output Generation - Generate human-readable explanations for each recommendation - Highlight reasons for penalization or promotion - Structure outputs in JSON format for integration	
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		Outcome: Transparent and interpretable recommendation results Task 9: Future Integration Readiness - Design interfaces to accept: - Job risk scores (Model 1) - Employer trust scores (Model 2) - Communication risk scores (Model 3) - Ensure modular replacement of simulated risk with real model outputs Outcome: Parallel-safe component ready for full system integration	
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Group Details:

	Student Name	Student Registration Number	Signature
1	Kajanthan U	IT22224002	U. Kajanthan
2	Santhosh S	IT22113740	S. Santhosh
3	Bandara H M N T	IT22251664	H. M. N. T. Bandara
4	Methmi N P C S	IT22188618	M. N. P. C. S. Methmi

**10. This part is to be filled by the
Supervisor and the Co-supervisor of the Project.**

- a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes ☒ No ☐

- b) The proposed topic exhibits novelty.

Yes ☒ No ☐

- c) The student group can successfully execute the proposed project.

Yes ☒ No ☐

- d) The proposed sub-objectives reflect the students' areas of specialization.

Yes ☒ No ☐

- e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor	Ms	Thilini	Jayalath	
Co-Supervisor	Ms	Karthiga	Rajendran	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				